

AI ASSISTANT CODING ASSIGNMENT-14.2

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Batch: 33

Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS Generation Lab Objectives

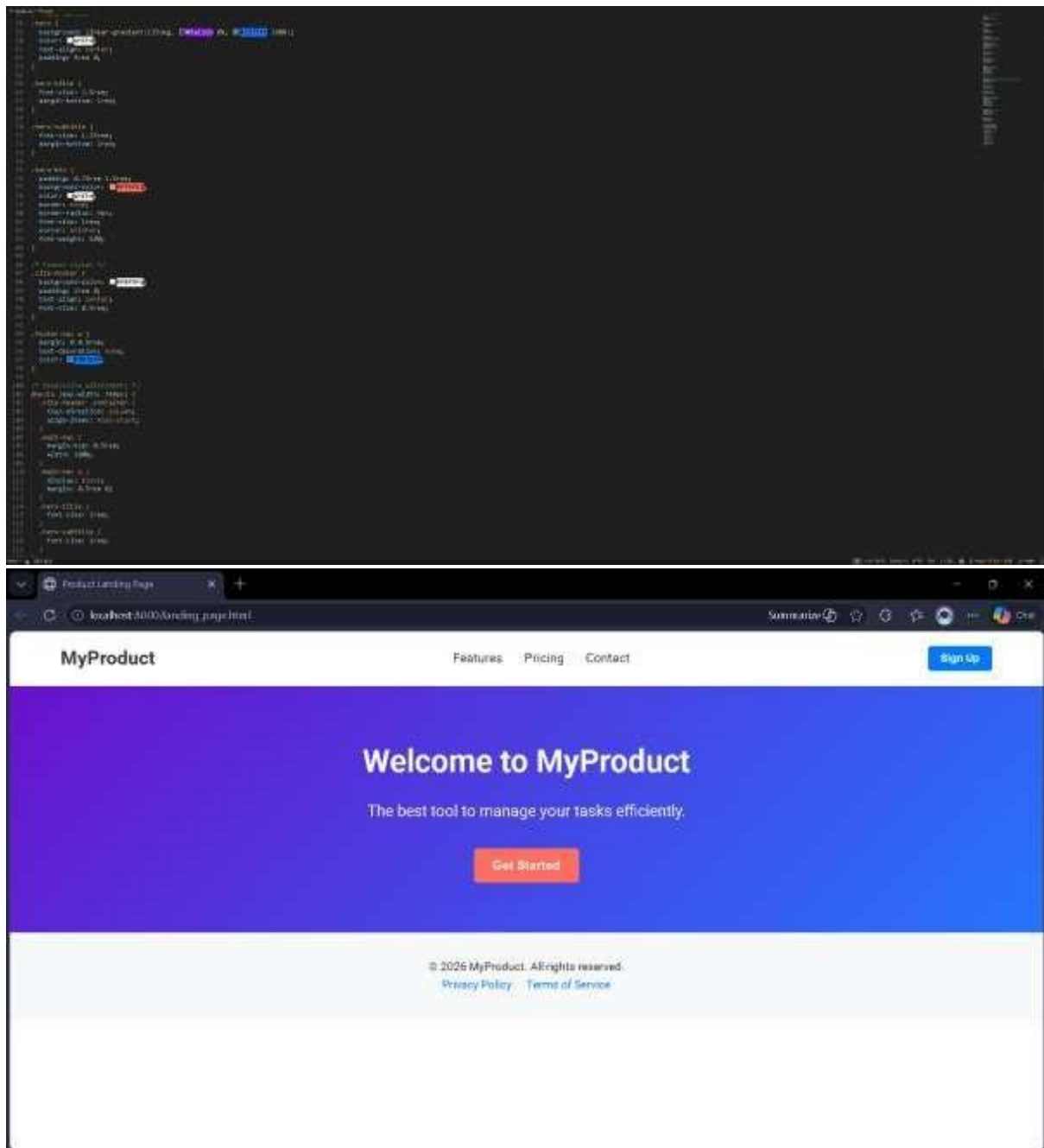
- Design functional, visually appealing web applications using HTML, CSS, and JavaScript with AI assistance.
- Apply responsive, accessible, and interactive design principles.
- Create practical UI components for real-world web applications.
- Use AI to optimize layout, UX, and performance.

Task Description 1 – (Landing Page Design)

- Create a simple landing page for a product or service. Include a header, hero section, and footer.

Requirements:

- Use HTML and CSS to structure the page.
- Use AI to suggest layout alignment and typography.
- Ensure the page is responsive across devices.



Task Description 2 – (Theme Toggle Feature)

- Provide a button to switch between light and dark themes..

Requirements:

- Use JavaScript to toggle CSS classes.
- Use AI assistance to manage color contrast and accessibility.
- Ensure theme preference applies instantly on click.

```

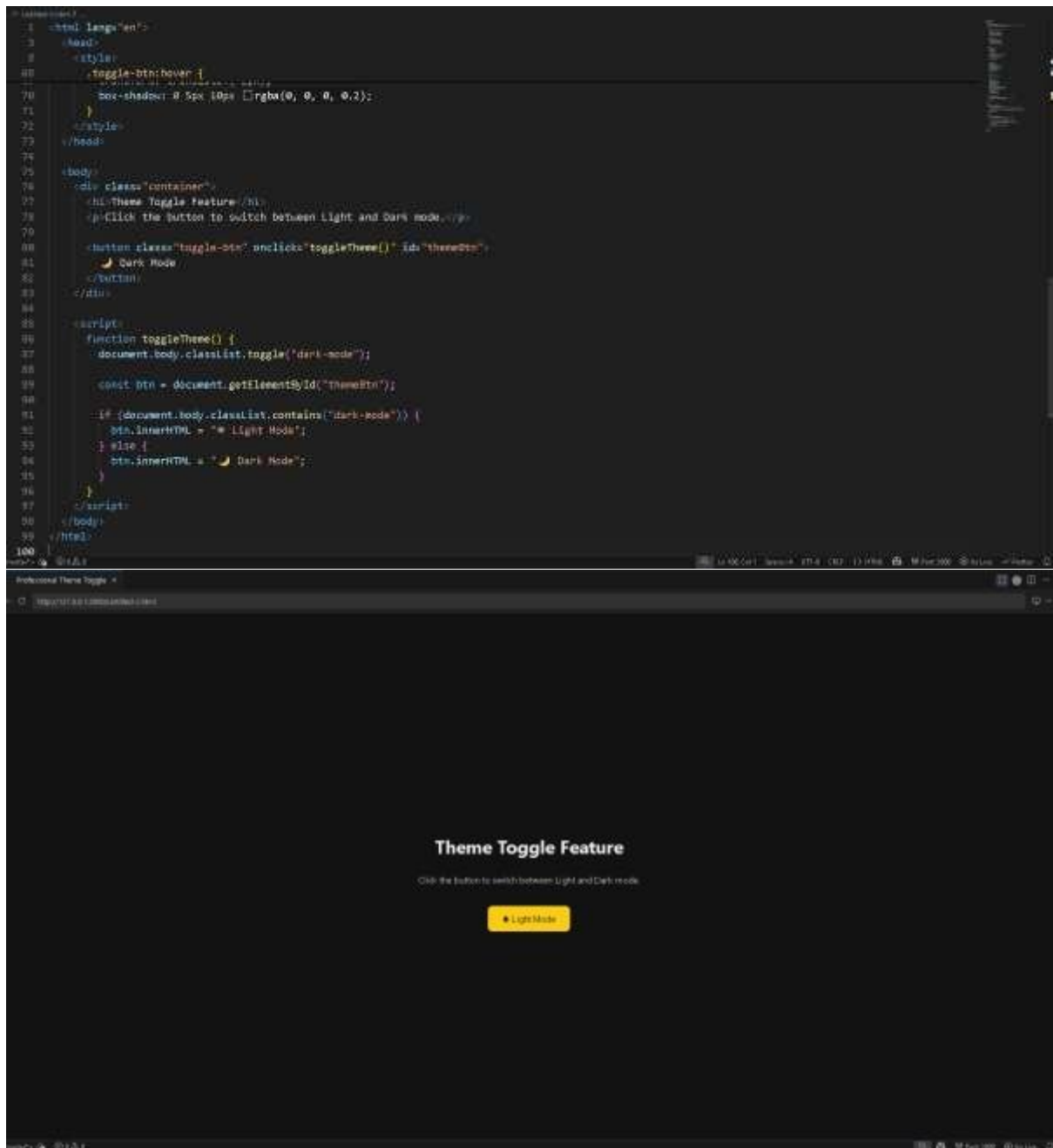
1 <!doctype html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6   <title>Professional Theme Toggle</title>
7
8   <style>
9     /* CSS Variables for Themes */
10    :root {
11      --bg-color: #ffffff;
12      --text-color: #212121;
13      --button-bg: #21863b;
14      --button-text: #ffffff;
15    }
16
17    body.dark-mode {
18      --bg-color: #121212;
19      --text-color: #f5f5f5;
20      --button-bg: #f0cc00;
21      --button-text: #000000;
22    }
23
24    /* Page Styling */
25    body {
26      margin: 0;
27      font-family:
28        Segoe UI,
29        Arial,
30        sans-serif;
31      background: var(--bg-color);
32      color: var(--text-color);
33      transition:
34        background 0.4s,
35        color 0.4s;
36    }
37  </style>
38 </head>

```

```

39 <body>
40   <div>
41     <h1>
42       Professional Theme Toggle
43     </h1>
44
45     <button>
46       Toggle Theme
47     </button>
48   </div>
49 </body>
50 </html>

```



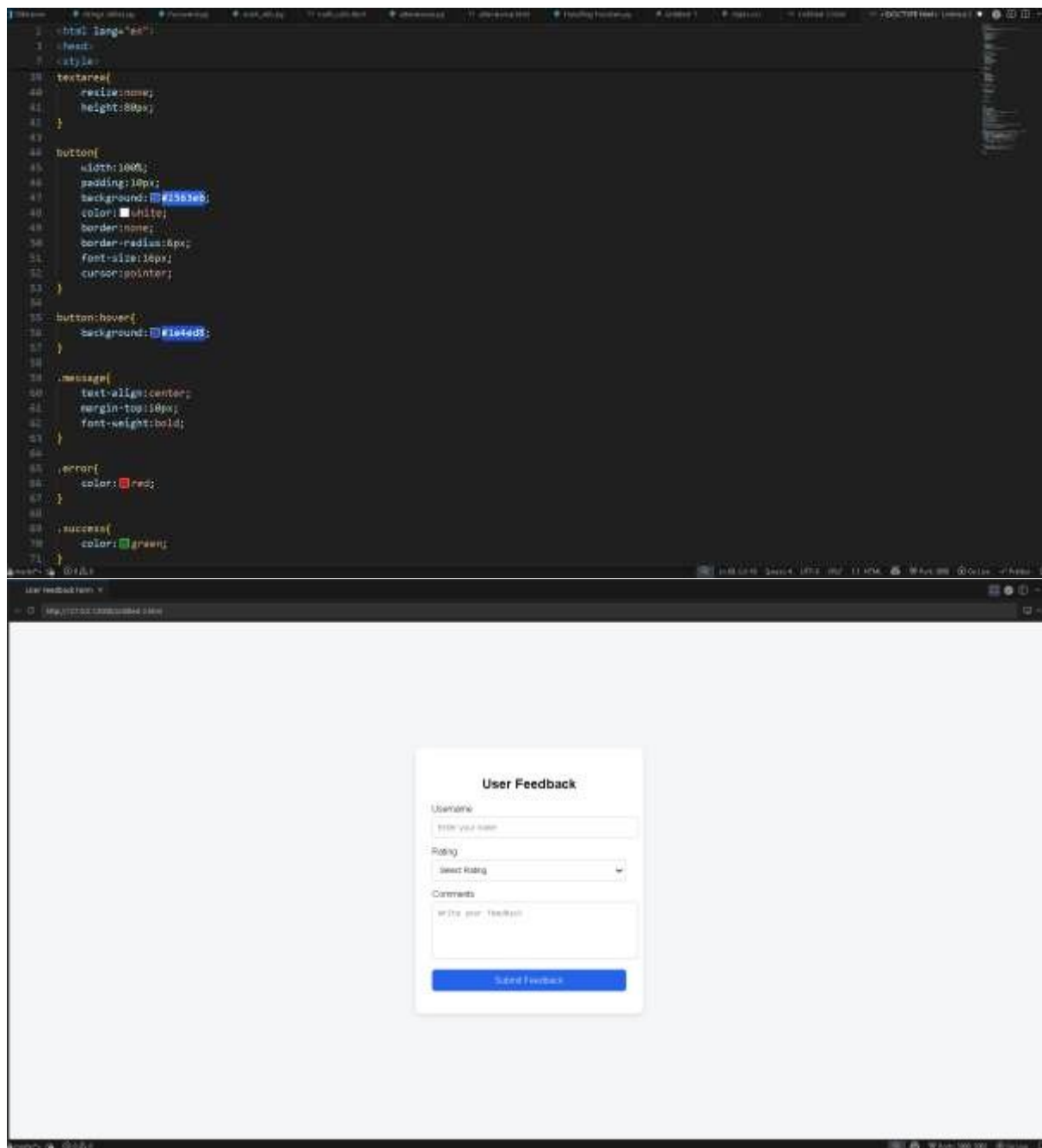
Task Description 3 – (User Feedback Form)

- Create a feedback form with fields: Username, Rating, Comments.

Requirements:

- Use JavaScript to validate required fields.
- Display success or error messages dynamically.
- Use AI to improve form usability and layout.

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>User Feedback Form</title>
7 <style>
8 body{
9   font-family: Arial, sans-serif;
10  background-color: #f0f0f0;
11  display: flex;
12  justify-content: center;
13  align-items: center;
14  height: 100vh;
15  margin: 0;
16 }
17
18 .container{
19   background: #ffffff;
20   padding: 20px;
21   border-radius: 10px;
22   width: 500px;
23   box-shadow: 0 4px 10px #000000;
24 }
25
26 h2{
27   text-align: center;
28   margin-bottom: 10px;
29 }
30
31 input, select, textarea{
32   width: 100%;
33   padding: 10px;
34   margin-top: 5px;
35   margin-bottom: 10px;
36   border: 1px solid #ccc;
37   border-radius: 5px;
38   font-size: 14px;
39 }
40
41 button{
42   width: 100%;
43   padding: 10px;
44   background-color: #000000;
45   color: #ffffff;
46   border: none;
47   border-radius: 5px;
48   font-size: 14px;
49   cursor: pointer;
50 }
51
52 #message{
53   margin-top: 10px;
54   padding: 10px;
55   border-radius: 5px;
56 }
57
58 .message-error{
59   background-color: #ffcccc;
60 }
61
62 .message-success{
63   background-color: #ccffcc;
64 }
65
66 @media (max-width: 480px){
67   .container{
68     width: 100%;
69     padding: 10px;
70   }
71 }
72
73 </style>
74 </head>
75 <body>
76 <div class="container">
77 <form id="FeedbackForm">
78 <div>
79 <input type="text" value="Username" />
80 </div>
81 <div>
82 <input type="text" value="Email" />
83 </div>
84 <div>
85 <select id="rating">
86 <option value="1">Very Good</option>
87 <option value="2">Excellent</option>
88 </select>
89 </div>
90 <div>
91 <label>Comments</label>
92 <textarea id="comments" placeholder="Write your feedback" />
93 </div>
94 <button type="submit">Submit Feedback</button>
95 </form>
96 <div id="message" class="message"></div>
97 </div>
98 <script>
99 document.getElementById("FeedbackForm").addEventListener("submit", function(e){
100   e.preventDefault();
101   let username = document.getElementById("username").value.trim();
102   let email = document.getElementById("email").value.trim();
103   let rating = document.getElementById("rating").value;
104   let comments = document.getElementById("comments").value.trim();
105   let message = document.getElementById("message");
106   if(username === "" || email === "" || rating === "" || comments === ""){
107     message.textContent = "Please fill all required fields.";
108     message.className = "message-error";
109   }
110   else{
111     message.textContent = "Feedback submitted successfully!";
112     message.className = "message-success";
113     document.getElementById("FeedbackForm").reset();
114   }
115 });
116 </script>
117 </body>
118 </html>
```



Task Description - 4 (Image Gallery with JavaScript)

- Task: Display a small image gallery on a web page.

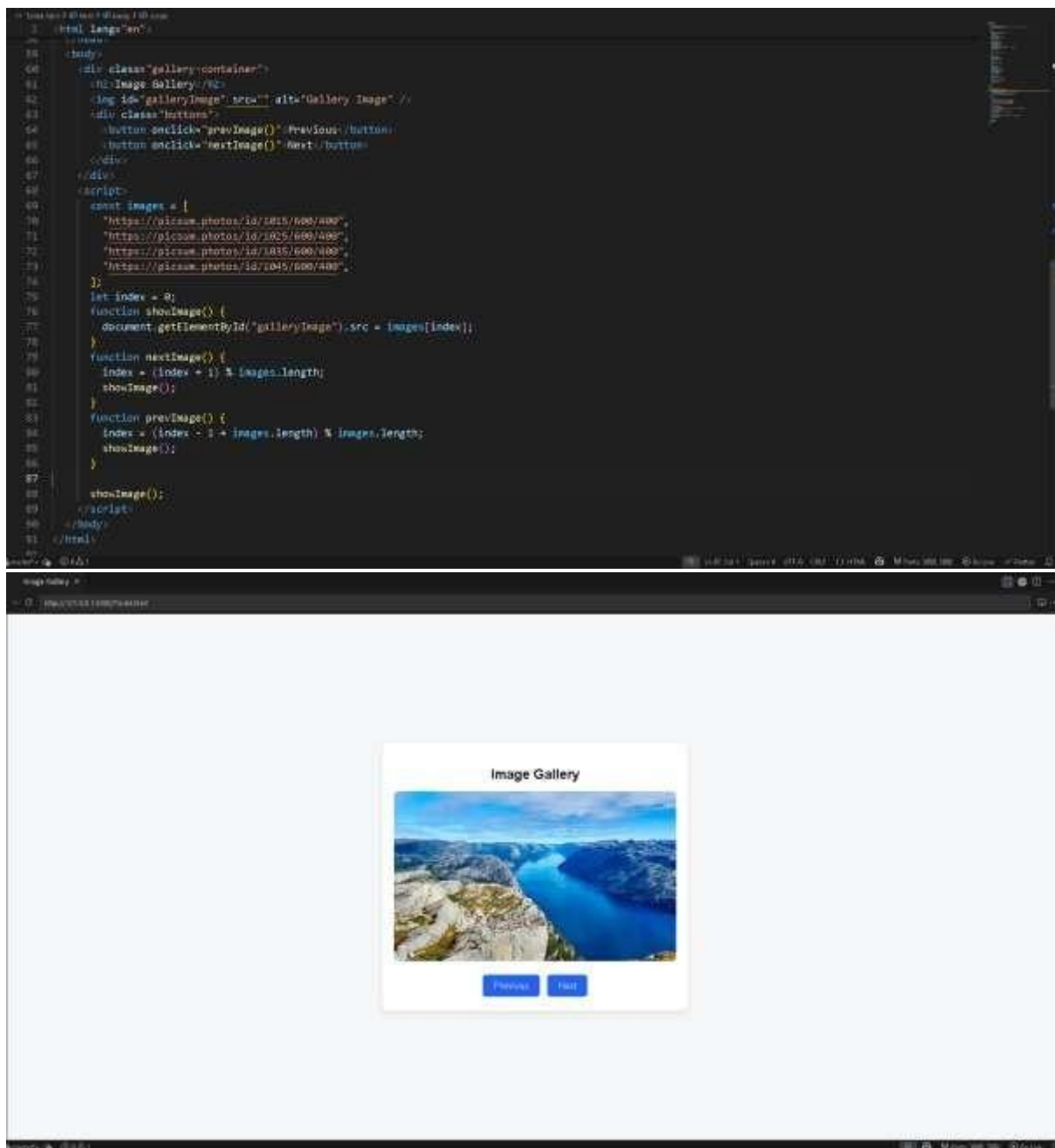
Requirements:

- Use JavaScript to dynamically load images.
- Add next/previous navigation buttons.
- Use AI to optimize layout responsiveness.

```

1 <!doctype html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8" />
5 <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6 <title Image Gallery />
7 <style>
8   body {
9     font-family: Arial, sans-serif;
10    background-color: #f4f6f8;
11    margin: 0;
12    display: flex;
13    justify-content: center;
14    align-items: center;
15    height: 100vh;
16  }
17  .gallery-container {
18    text-align: center;
19    background-color: white;
20    padding: 20px;
21    border-radius: 10px;
22    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
23    max-width: 500px;
24    width: 90%;
25  }
26  h2 {
27    margin-bottom: 15px;
28  }
29  img {
30    width: 100%;
31    height: 300px;
32    object-fit: cover;
33    border-radius: 5px;
34  }
35  .buttons {
36    margin-top: 15px;
37  }
38  button {
39    padding: 10px 20px;
40    margin: 5px;
41    border: none;
42    border-radius: 5px;
43    background-color: #f2f3f8;
44    color: white;
45    font-size: 16px;
46    cursor: pointer;
47  }
48  button:hover {
49    background-color: #f4f6f8;
50  }
51  @media (max-width: 600px) {
52    img {
53      height: 200px;
54    }
55  }
56 </style>
57 </head>
58 <body>
59 <div class="gallery-container">
60 <h2>Image Gallery />
61 <img id="galleryImage" src="" alt="Gallery Image" />
62 <div class="buttons">
63 <button onclick="previousImage()">Previous />
64 <button onclick="nextImage()">Next />
65 </div>
66 </div>
67 </body>
68 </html>

```

Task Description - 5: (Notification Banner)

- Display a notification banner at the top of the page.

Requirements:

- Use JavaScript to show and hide the banner.
- Style the banner using CSS.
- Use AI to ensure smooth transitions and clean UI behavior.

```

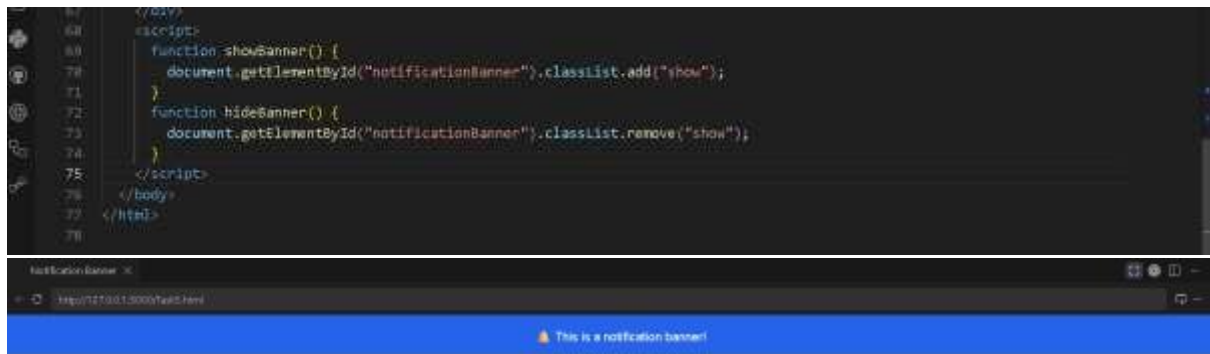
1 <!doctype html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8" />
5 <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6 <title>Notification Banner</title>
7 <style>
8 /* Page styling */
9 body {
10     margin: 0;
11     font-family: Arial, sans-serif;
12 }
13 /* Notification banner */
14 .banner {
15     position: fixed;
16     top: -100px;
17     left: 0;
18     width: 100%;
19     background: #2556a8;
20     color: white;
21     text-align: center;
22     padding: 15px;
23     font-size: 16px;
24     transition: top 0.5s ease;
25 }
26 /* when banner is visible */
27 .banner.show {
28     top: 0;
29 }
30 /* Close button */
31 .close-btn {
32     position: absolute;
33     right: 10px;
34     top: 10px;
35     cursor: pointer;

```

```

36 }
37
38 /* Content area */
39 .container {
40     padding: 10px 20px;
41     text-align: center;
42 }
43
44 button {
45     padding: 10px 20px;
46     font-size: 16px;
47     border: none;
48     border-radius: 5px;
49     background: #2556a8;
50     color: white;
51     cursor: pointer;
52 }
53
54 button:hover {
55     background: #1e4d8d;
56 }
57 </style>
58 </head>
59 <body>
60 <div id="notificationBanner" class="banner">
61     ⚠ This is a notification banner!
62     <span class="close-btn" onclick="hideBanner()">✖</span>
63 </div>
64 <div class="container">
65     <p>Notification Banner Example</p>
66     <p>Click the button to display the notification.</p>
67     <button onclick="showBanner()">Show Notification</button>
68 </div>
69 </body>
70 </html>

```



Notification Banner Example

Click the button to display the notification.

Show Notification